Unit 3

Lesson 2

Beyond Imagination

Wave Hello!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Understand what machine learning is.
- Train a computer to tell hand gestures apart using PictoBlox.
- Connect each hand sign to a different sound or music note.
- Understand what a loop is and use repeat blocks to make actions happen again
- Learn how to use repeat blocks to make actions happen again and again in your code.
- Use if-else blocks to help your program make choices based on what's happening.

Challenge questions of the lesson

1. What is Machine Learning?
2. Can a computer tell which sign I'm showing and play a matching sound—just like a musical instrument?
3. How can my program decide what to do based on which hand sign I show?

SEL Relation

- Self-Awareness
    - Students reflect on how their own body (e.g., hand gestures) can be used to interact with technology.
    - hey become more aware of how their physical actions (like showing a hand sign) are perceived by others—including computers.
- Self-Management
    - Training a model takes patience and care.
    - Students must consistently show signs, manage frustration when the model doesn't work perfectly, and persist through trial-and-error.
- Social Awareness
    - Using gestures to trigger actions helps students understand nonverbal communication.
    - They learn to appreciate how people who are nonverbal or have limited mobility might communicate using alternative methods like sign language.
- Relationship Skills
    - Students may collaborate in pairs or small teams to train the model, test code, and troubleshoot.
    - They practice communication, encouragement, and active listening when giving and receiving feedback.
- Responsible Decision-Making
    - Using if-else logic mimics real-life decision-making.
    - Students learn to program smart choices into their projects—and this mirrors how they make decisions based on what's happening around them.

Engage

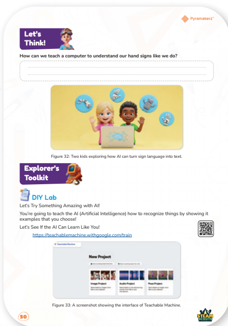
Teacher's Role:

- **Curiosity Builder:**
  Begin by asking:
  "What if a computer could recognize your hand signs like 'peace' or 'stop'?"
  Get students curious about how computers can learn from us.
- **Facilitator of Wonder:**
  Show a short demonstration or video of a computer reacting to different hand signs (can be from PictoBlox or real-world use cases like sign language readers).
- **Bridge to the Real World:**
  Prompt students to think about why we might want to teach computers to recognize gestures (e.g., for music, communication, games, accessibility).
- **Prompt for Prediction:**
  Ask:
  "What do you think happens behind the scenes when the computer 'learns' your hand?"
  Let them guess freely—no wrong answers at this stage.

Expected from Students:

- **Active Participation:**
  Students observe, ask questions, and express curiosity about how machine learning works.
- **Personal Connections:**
  Students relate the idea to their own experiences—like using hand gestures in daily life or using tech that reacts to movement (e.g., Kinect, Alexa, or games).
- **Prediction and Imagination:**
  Students suggest how a computer might "know" what your hand is doing and imagine what their own project might do.
- **SEL Tie-In:**
  Students reflect on how different gestures can communicate meaning—especially for people who use signs to communicate.

Parts of the books included.



Story

1. **Set the Scene with Energy and Curiosity**
   - Say:
     "Remember how we made our button-controlled device talk last time? What if we didn't need the button at all? What if... it could read our hand signs?"
   - Point to the device or image from the last session to remind students.
2. **Narrate the Story with Expression**
   - Read the dialogue aloud, using expressive voices for each character.
   - Pause between lines to show character emotions (e.g., "Adam (curious): 'You mean...?'").
3. **Act It Out**
   - Ask three volunteers to role-play as Ms. Sara, Adam, and Lila.
   - Encourage the rest of the class to watch and think:
     "What are they trying to solve now?"
4. **Ask a Wondering Question**
   - "Do you think a computer can see your hand and know what it means?"
   - "If you had to teach a computer what 'stop' means with your hand, how would you do it?"
5. **Connect to Today's Activity**
   - Say:
     "Just like Adam and Lila said, we'll teach the computer to recognize our hands today—and even make it do something fun like play a sound!"

Possible Strategies for Implementation:

- **Use Visual Prompts**
  Show simple images or real hand signs to spark predictions:
  "What might the computer think this means?"
- **Activate Prior Knowledge**
  Ask: "Have you used anything before that reacts when you move—like hand gestures in games or phones?"
- **Think-Pair-Share**
  Let students turn to a partner and finish this sentence:
  "I think the computer knows what my hand is doing because..."
- **Anchor in Real-World Use**
  Briefly mention how sign language, touchless tech, and gesture-based controls are used in real life—for inclusion, safety, and creativity.

- **Emotion Check-In**
  Ask students to show a gesture for how they feel about the challenge (e.g., excited, nervous, curious) and reflect together.

Let's Think!

Objective:

Students will begin to understand the basic idea of machine learning by thinking about how a computer can be trained to recognize and respond to human hand gestures.

Possible Correct Answers:

- "We can show the computer pictures of hand signs again and again until it learns what each one means."
- "We can teach it by practicing different signs in front of the camera."
- "We tell the computer: 'This is stop,' 'This is go,' and it tries to remember."
- "It watches and learns from us—like training a pet!"

Possible Strategies for Implementation:

1. **Use Everyday Comparisons**
   - Strategy: "Like a Learning Pet"
     - o Say: "Imagine you're teaching a puppy to sit. You say 'sit,' show the motion, and reward it. Computers learn a little like that too!"
     - o Follow-up: "So how would you teach a computer to know when you're doing 'thumbs up'?"
2. **Show a Quick Demonstration**
   - Strategy: "Train the Teacher!"
     - o Do a few hand gestures and ask students to guess what they mean.
     - o Ask: "Now, what if the computer had to learn these too—how would it do that?"
3. **Prompt with Visual Thinking**
   - Strategy: "Match the Sign to the Action"
     - o Show the icons under the question (robot hand, AI brain, camera, etc.) and ask:

   1. "What is this for?"
   2. "How does this help a computer learn?"

     - o Let students connect: "Ah, the camera helps it see!" or "That robot might be the one learning."
4. **Encourage Prediction and Imagination**
   - Strategy: "What If...?"

- o Ask: “What would happen if the computer got the sign wrong? How can we help it get better?”
- o Invite silly or creative guesses, then tie back to the need for training data and repeat practice.

Explorer’s Toolkit

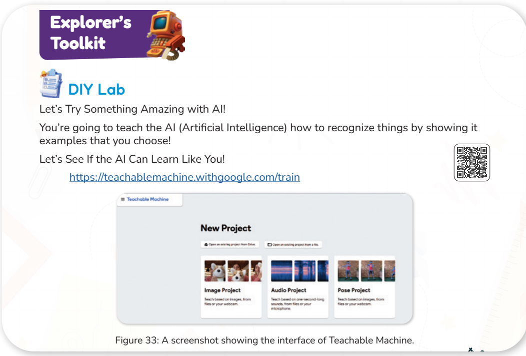
Teacher’s Role:

- Facilitator: Introduce the concept of machine learning through hands-on exploration.
- Technical Support: Help students access the Teachable Machine site and guide them through creating a "Pose Project."
- Encourager: Prompt students to try different gestures and test how accurate the model becomes with more training.
- Observer & Coach: Circulate, observe student progress, and ask probing questions like “Why do you think it’s not recognizing this sign?” or “What if we give it more examples?”

Expected From Students:

- Create a new Pose Project in Teachable Machine.
- Train two or more hand signs using their webcam.
- Observe how the model responds and test if it can tell the signs apart.
- Reflect on how more examples improve accuracy.

Parts of the books included


Explorer’s Toolkit

DIY Lab

Let’s Try Something Amazing with AI!

You’re going to teach the AI (Artificial Intelligence) how to recognize things by showing it examples that you choose!

Let’s See If the AI Can Learn Like You!

https://teachablemachine.withgoogle.com/train

Figure 33: A screenshot showing the interface of Teachable Machine.

DIY Lab

Objective:

Students will understand how machine learning works by training a computer to recognize hand gestures through real-time examples using Teachable Machine.

Materials Needed:

- Laptops or tablets with internet access

- Webcam (built-in or external)
- Access to Teachable Machine
- Printed labels or post-its (optional, for gesture cues)

Activity link: https://teachablemachine.withgoogle.com/train

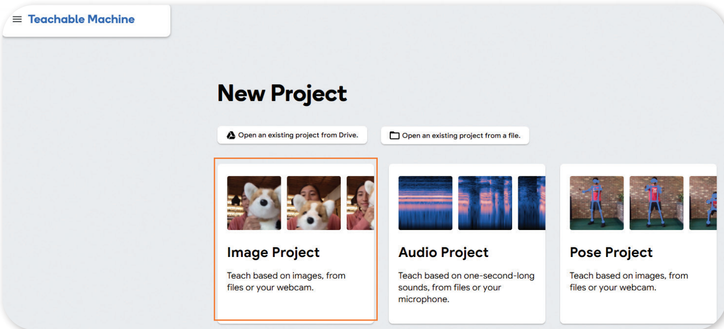
Activity Instructions:

1. **Introduce Machine Learning:**

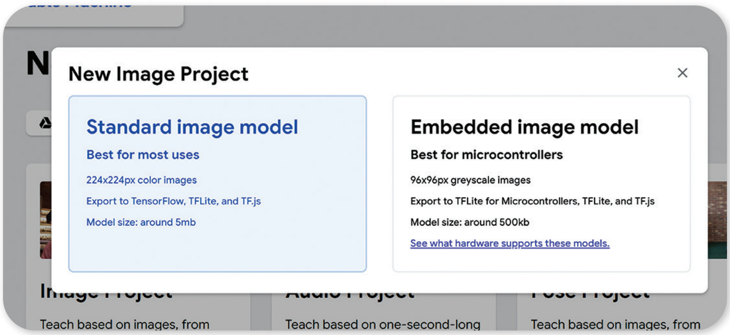
- Explain: "You're going to teach a computer, not with code—but by showing it examples!"

2. **Demonstrate Setup:**

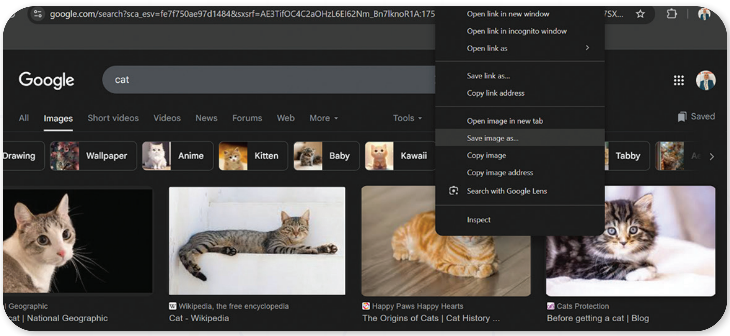
- Open the Teachable Machine site and select image Project.



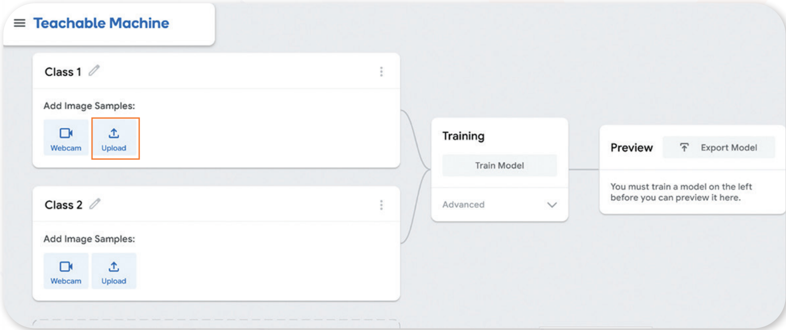
- Click on new image project.



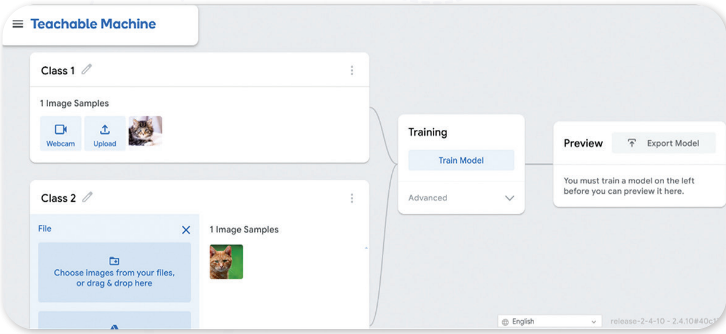
- Work on a cat recognition model by downloading 10 to 20 images of various cats from Google.



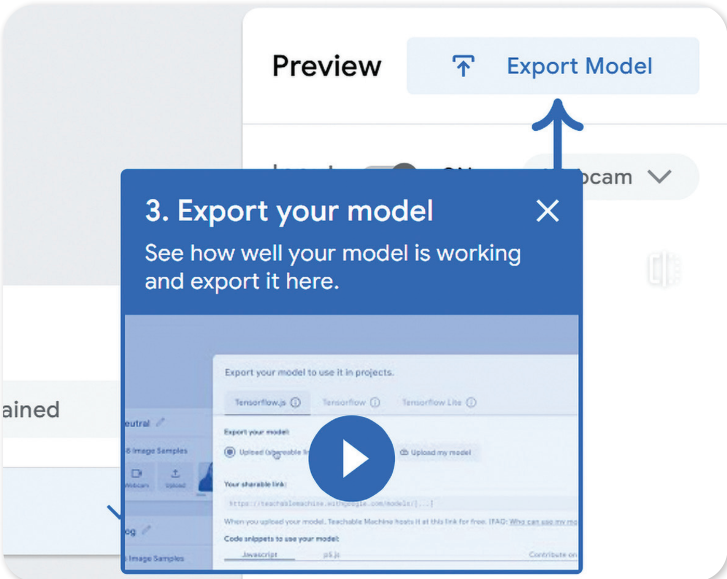
- Click on class1, upload and start to upload the first photo, repeat the step 20 times to upload all photos.



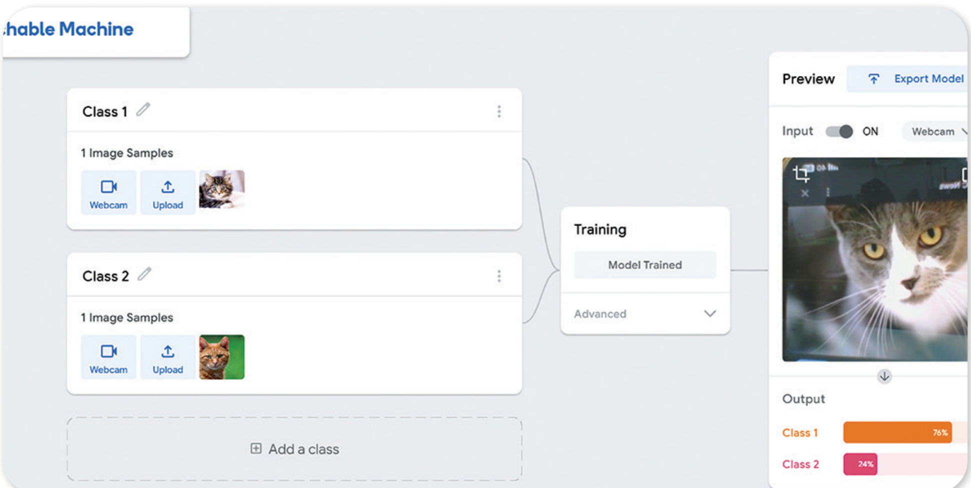
- Click on training model.



- Click on export model to test it.



- Open another different cat photo and test its relative to which class.



3. **Discuss Model Accuracy:**

- Ask: “Did it guess correctly? What might help it improve?”
- Encourage retesting with added examples.

Possible Strategies for Implementation:

1. **Pair Programming.**

- Let students work in pairs—one acts out gestures while the other manages the computer.
- Fosters teamwork and splits focus between training and observation.

2. **Gesture Cards.**

- Provide pre-made gesture cards (e.g., thumbs up, stop hand, peace sign) to reduce confusion and ensure clarity in what to teach the computer.

3. **Time Checkpoints**

- Use a 10-minute timer for each phase (recording gestures, training, testing), keeping the activity structured and paced.

4. **Error Exploration**

- Celebrate mistakes. Ask: “Why do you think it got confused?” to help kids understand the need for more data.

Explain

Teacher’s Role:

- **Introduction Leader:** Set the stage by reviewing what students did in the DIY lab and introduce the purpose of the video—understanding how the computer “learns.”
- **Guide for Focus:** Before the video, guide students to focus on specific ideas (e.g., training with examples, how computers make decisions).
- **Discussion Facilitator:** After the video, lead a discussion on what machine learning means and how it’s used in their own project.
- **Clarifier:** Be ready to pause the video to explain technical terms or show real examples students can relate to.

Expected From Students:

- Listen attentively to the video and look for examples of how the computer learns from images or input.
- Take note (mentally or on paper) of new words or important ideas like “training data,” “prediction,” or “accuracy.”
- Share reflections and connections to their own experience using Teachable Machine or programming the hand gesture model.

Parts of the books included.

Screen Lessons

Watch the video learn more machine learning.

https://www.youtube.com/watch?v=ZdaVXyan8GM

Figure 34: A boy learning about machine learning.

Screen Lessons

Objective:

Students will develop an understanding of how machine learning works by watching an introductory video. They will identify key concepts such as how computers learn from examples and how this relates to training a model with data.

Video link: https://www.youtube.com/watch?v=ZdaVXyan8GM

Possible Strategies for implementation:

1. **Pre-Watching: Set a Purpose**
   - Strategy: “Question Before Watching”
     Ask:
     - “What do you think machine learning means?”
     - “How do you think a computer can recognize a cat or a hand sign?”
     - Write answers on the board. Tell students they’ll be finding the answers in the video.
2. **During Watching: Active Engagement**
   - Strategy: “Pause and Prompt”
     Pause the video at key points to ask:
     - “What did the computer learn from the images?”
     - “Why does the model need more examples?”
     - Let students turn and talk to a partner for short moments to keep engagement high.
3. **Post-Watching: Discussion and Reflection**
   - Strategy 1: “Machine Learner Check-In”
     Ask: “What surprised you most about how the computer learns?”
   - Strategy 2: “Connection to Our Work”
     Ask: “How is this similar to what we did in Teachable Machine?”
   - Strategy 3: “Explain Like a Teacher”
     Challenge students to explain what machine learning is to a younger student.

Elaborate

Teacher's Role

- **Game Facilitator:** Distribute sign language cards, explain the rules, and ensure students stay silent during gestures. Monitor and encourage participation.
- **Supportive Coach:** Help students observe carefully, guess meaningfully, and build empathy for non-verbal communication.
- **Tech Guide:** Assist students in launching PictoBlox, navigating the sound library, and troubleshooting any setup issues.
- **Creativity Encourager:** Motivate students to explore different types of sounds—funny, musical, or realistic—and think about which signs they might match.
- **Safety Monitor:** Make sure gestures and group interaction are respectful and inclusive.

Expected From Students

- **Non-Verbal Communicators:** Act out the sign on their card using only hand gestures and facial expressions—no words!
- **Guessing Participants:** Watch peers closely and guess the meaning of their signs.
- **Critical Thinkers:** Reflect on what made each sign easy or hard to understand and relate it to accessibility.
- **Digital Explorers:** Start a PictoBlox project, go to the Sounds tab, and begin building a personal sound library.
- **Creative Coders:** Choose or record at least 3–5 distinct sounds and start thinking about how to assign each to a different hand sign in the next phase.

Parts of the books included.


Screen Lessons
Watch the video learn more machine learning.
https://www.youtube.com/watch?v=ZdaVXyan8GM

Figure 34: A boy learning about machine learning.

Assessment

Apprentice
Let's Play: Guess the Sign!
You will each have a set of sign language cards.
1. Look at the sign on your card.
2. Do the sign with your hands — no speaking!
3. Let your classmates guess what it means.
4. Take turns and see how many signs your friends can guess!

HELLO
Raise your hand and move it side to side.

After you are done let's play this game and solve the maze with our hand gestures.

Down Right Up Left

STEAM SCOUTS 51


Pyramakerz®

Hello friends!

Machine learning is when a computer learns how to do something **by looking at examples,** just like we do!

Figure 35: Machine learning.

Builder
Start a new project in PictoBlox. Then, click on the "Sounds" tab. You can choose sounds from the library or upload your own recorded sounds.

You're creating your sound library — you'll match each sound to a hand sign later!

Figure 36: A PictoBlox interface showing the "Sounds" tab.

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Apprentice

Objective:

Students will explore the basics of sign language and hand gestures through a silent guessing game, helping them build empathy, observation skills, and a foundation for gesture recognition in computing.

Activity Links:

Sign Language Resources Folder: https://drive.google.com/drive/folders/1TOhRwrcSQRjxsyMT6PytfJKjonbaFapJ

Maze game resources game: https://drive.google.com/drive/folders/1m1OeiuLHlaSG0e4qUJNLuds5AoUm8kPy

Activity Instructions:

1. **Prepare Materials**

- Download and print sign language cards from the Google Drive link.
- project the hand maze game.

2. **Introduce the Activity**

- Explain the goal: Students will guess hand signs using only gestures—no speaking allowed!
- Remind students this is about empathy and understanding others who use gestures to communicate.

3. **Game Time – Let's Guess the Sign!**

- Hand out one card per student or pair.
- Give students 30 seconds to practice the sign silently.
- One by one, students take turns acting out their signs while classmates guess.
- After all turns, reflect: What made some signs easier/harder to guess?

4. **Maze Challenge**

- Using the hand gesture directions (up, down, left, right), students solve the maze as a group or in pairs.

Possible Strategies for Implementation:

- **Model First**
  - Pick one card and act out the gesture. Let students guess.
  - Reinforce the rule: "No speaking during gestures!"
- **Turn & Talk**
  - Before guessing, give students 10 seconds to whisper guesses to a partner to build confidence.
- **Think-Aloud Reflection**
  - After the game, ask:

"How do people who are deaf or non-verbal share ideas like this every day?"

- **Connection to Coding**
  - o Briefly highlight how computers will "guess" signs too using image training—just like we did.
- **Anchor Visuals**
  - o Keep a sign key or poster visible during the activity for reference and reinforcement.

Builder

Objective:

Students will create a personalized sound library in PictoBlox by selecting or uploading different sound clips. These sounds will later be linked to hand signs as part of their talking device program.

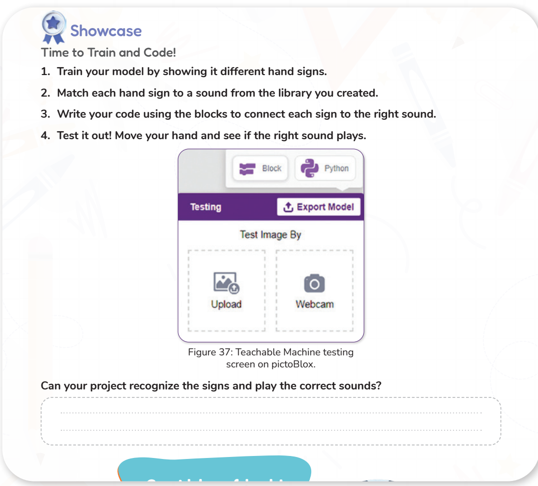
Teacher's Role:

- **Support Setup:** Help students access their trained model and guide them in integrating it into PictoBlox.
- **Facilitate Debugging:** Provide hints and encouragement if blocks aren't working as expected.
- **Encourage Reflection:** Ask guiding questions such as "Why do you think the wrong sound played?" or "What could we adjust to make the model better?"
- **Celebrate Success:** Acknowledge both successful outputs and thoughtful troubleshooting.

Expected From Students:

- Successfully train a model with at least 2–4 hand signs.
- Link each hand sign to a specific sound from their custom library.
- Use if-else blocks and loops to connect the model to the sound.
- Test their system and reflect on whether it works correctly.
- Show perseverance in improving the model and code if something fails.

Parts of the books included.


Showcase

Time to Train and Code!

1. Train your model by showing it different hand signs.
2. Match each hand sign to a sound from the library you created.
3. Write your code using the blocks to connect each sign to the right sound.
4. Test it out! Move your hand and see if the right sound plays.

Figure 37: Teachable Machine testing screen on pictoBlox.

Can your project recognize the signs and play the correct sounds?

Showcase

Objective:

Students will create a personalized sound library in PictoBlox by selecting or uploading different sound clips. These sounds will later be linked to hand signs as part of their talking device program.

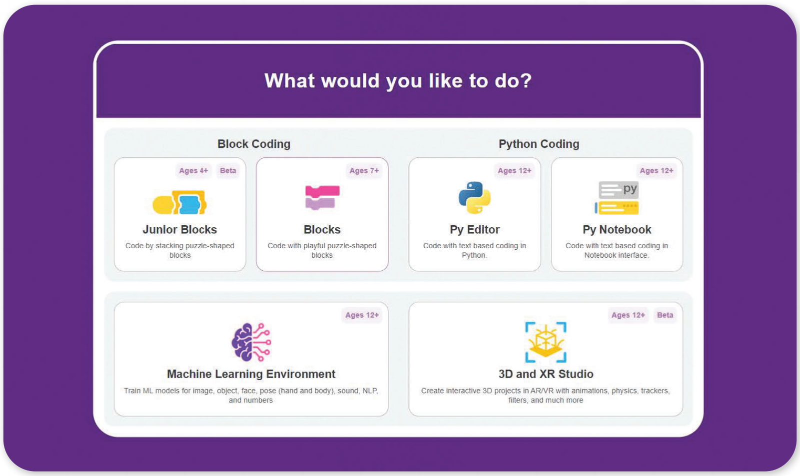
Activity Resources:

https://drive.google.com/file/d/1gqUl1ZaggL7ulH-ex3OmnG-DtDKK268n/view?usp=drive_link

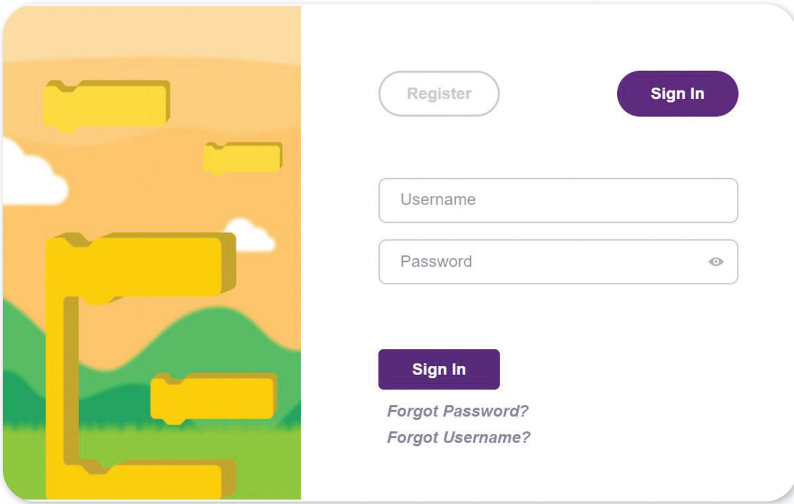
Activity Instructions:

Open PictoBlox:

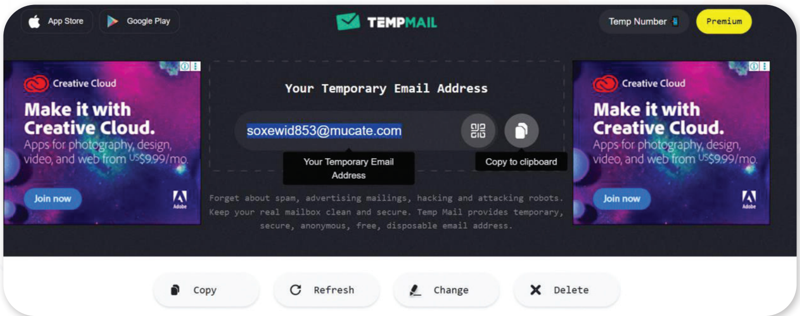
Guide students to launch PictoBlox and start a new project.



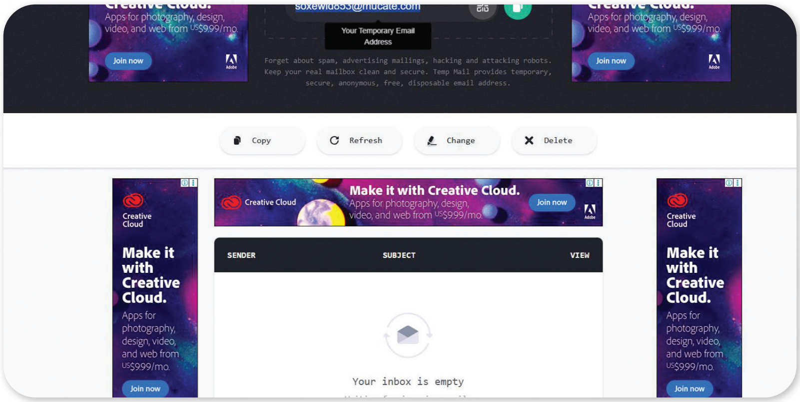
Sign in in pictoblox to use AI features:

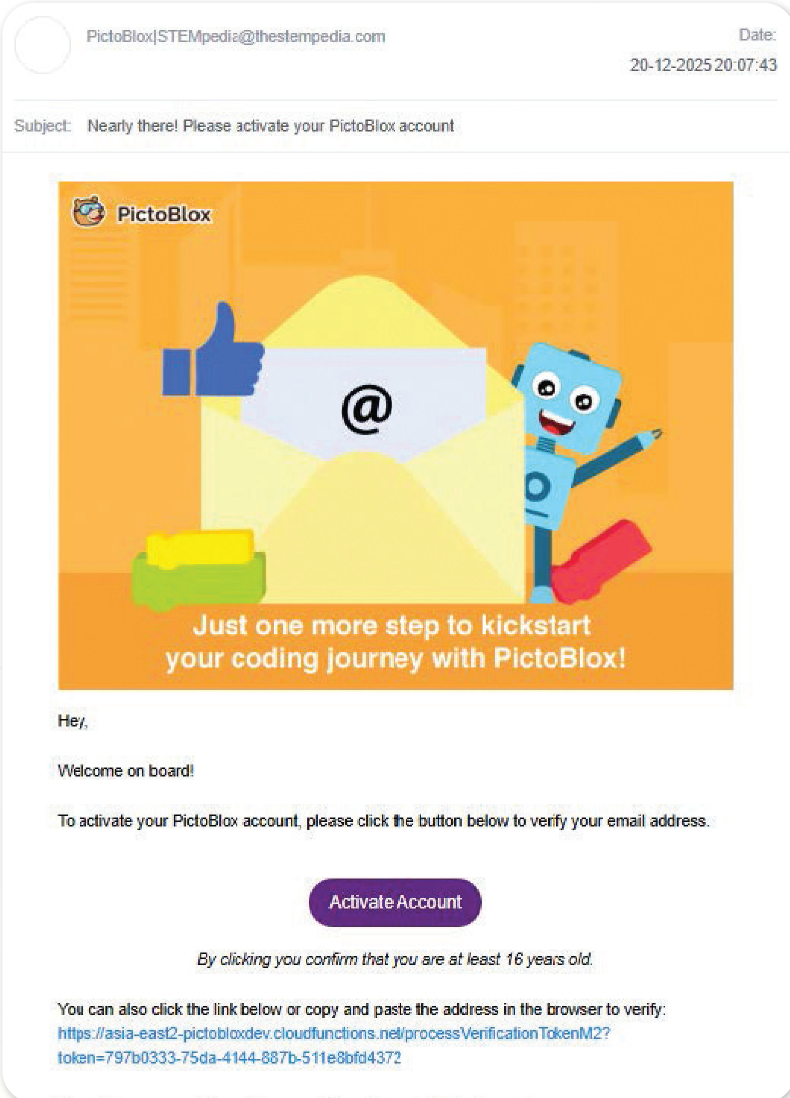


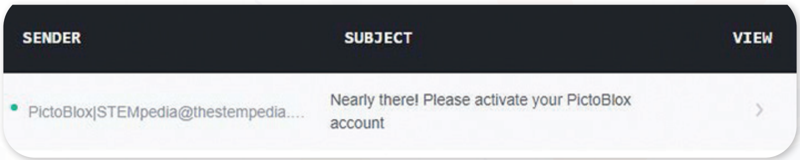
Note: Each account has a limited credit, and once it is used up, the AI features stop working. In this case, we can use Temp Mail (https://temp-mail.org/) to quickly generate an email address for signing in.



After signing up, you will receive a verification email, which can be found in the Temp Mail inbox.



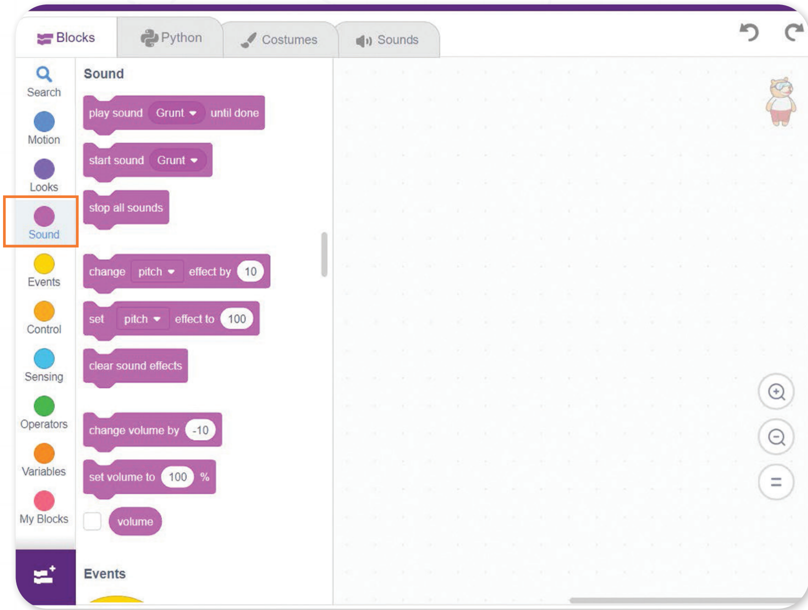




Back to pictoblox:

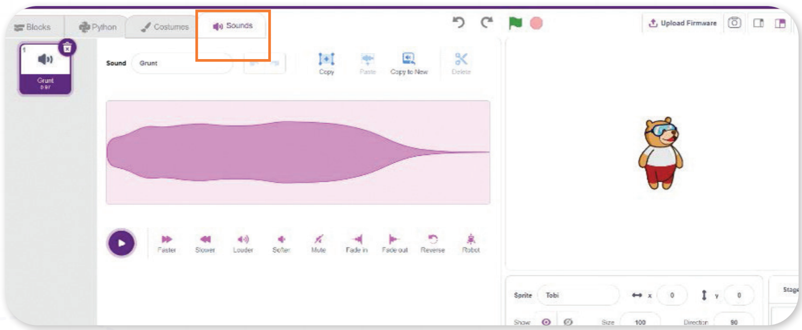
Navigate to the Sounds Tab:

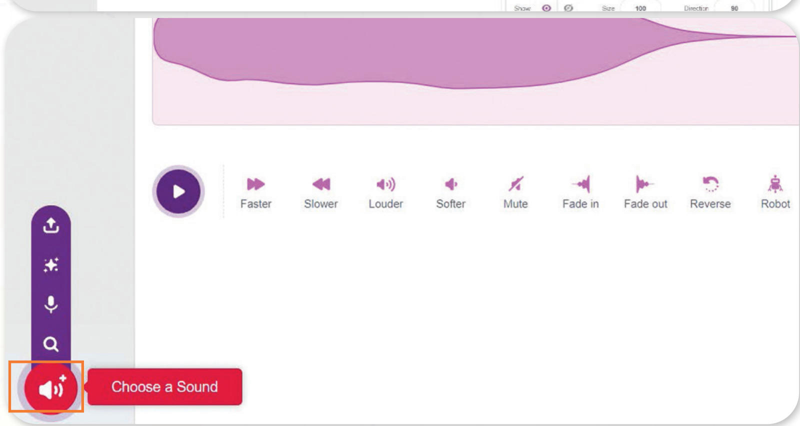
- Have students click on the “Sounds” tab.

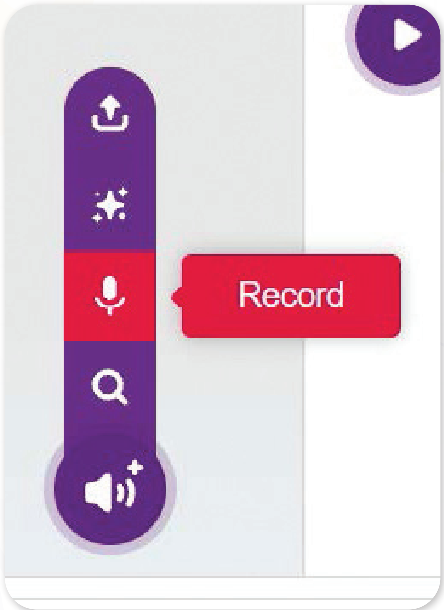


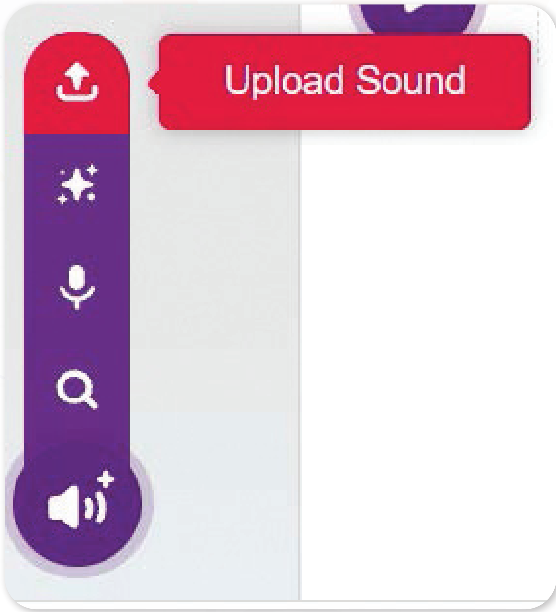
They can choose from:

- Built-in sound effects.
- Voice clips from the library.
- Upload their own recorded sounds.









Evaluation Rubric

Criteria	Excellent (4 pts)	Good (3 pts)	Needs Work (2 pts)	Try Again (1 pt)
Model Training	Taught the computer 3+ clear hand signs	Taught 2 clear hand signs	Only 1 sign trained or some signs unclear	Didn't finish training the model
Sound Matching	Each sign plays the correct sound	Most signs play the right sound	Some sounds don't match the sign	No sounds play correctly
Code Blocks Used	Used repeat and if–else blocks correctly	Used most blocks correctly	Some blocks used incorrectly	Didn't use blocks or code doesn't run
Testing & Fixing	Found mistakes and fixed them	Found and fixed some mistakes	Tried testing, but didn't fix problems	Didn't test or fix anything
Creativity & Effort	Added fun sounds or made it playful	Added some nice ideas	Simple but works	Did very little work

It’s time to move to the improve part and add this

Now I Can

Objective:

Students will reflect on their learning by identifying key coding and machine learning concepts they’ve practiced, such as gesture recognition, sound mapping, loop and conditional blocks, and understanding how a computer can be trained using examples.

Activity Instructions:

- **Display the "Now I Can" list on the board or hand out printed copies.**
- **Read each statement aloud with students and pause after each to ask:**
  - o “Can you give an example of when you did this in today’s activity?”
  - o “Which one was your favorite to learn?”
- Let students put a check next to each statement they feel confident about.
- **For further engagement:**
  - o Have students pair up and explain one skill to a classmate using their own words.
  - o Ask students to draw or write one thing they’re proud of learning.

Relation to Standard

1. NGSS – Next Generation Science Standards

3-PS2-4:
Define a simple design problem that can be solved by applying scientific ideas about magnets (used when building circuits and interacting with sensors).

3-5-ETS1-1:
Define a simple design problem reflecting a need or a want that includes criteria for success and constraints on materials, time, or cost (related to designing a protective case for the talking device).

3-5-ETS1-2/3:
Generate and compare multiple solutions, plan and carry out fair tests.

2. **ISTE Standards for Students**

1.4 Innovative Designer:
Students use a deliberate design process for generating ideas, testing theories, and creating innovative artifacts.

1.5 Computational Thinker:
Students understand and apply logical reasoning and use pattern recognition (e.g., training models, coding hand gestures).

1.1 Empowered Learner:
Students take ownership of learning goals by reflecting on and showcasing what they learned (seen in "Now I Can" and rubric parts).

3. **CASEL SEL Competencies**

Self-Awareness & Confidence: Students reflect on their ideas and learning in "Now I Can" and design tasks.

Relationship Skills:
Collaborative games and peer feedback in guessing signs and team projects.

Responsible Decision-Making:
Students consider user needs (e.g., safety, ease of use) when designing smart devices.
CCSS-Math – Common Core State Standards for Math (Grade 3)

3.MD.C.5–7:
Understand area as an attribute of plane figures and relate area to multiplication and addition.

MP1 & MP4:
Make sense of problems and persevere in solving them; model with mathematics (used in case measurements and area calculations).

Final Scene

Objective:

Students will celebrate the success of their smart talking device and reflect on how teamwork, design thinking, and empathy helped them solve a real-world communication problem. They will recognize the value of invention in improving lives.

How the Teacher Should Do It:

1. **Set the Stage with Excitement**
   - Say: "The big moment is here! Let's imagine we're all in the classroom watching Omar test the talking device. What do you think will happen?"
   - Project the final scene on the board or read it aloud with dramatic voice shifts for each character.
2. **Act it Out with Expression**
   - Assign student volunteers to act as Ms. Sara, Adam, Lila, and Omar.

- Encourage students to use emotion and gestures (e.g., big smiles, pressing a pretend button, cheering).
- Optionally, use a real button circuit built earlier to simulate the sound.

3. **Highlight the Engineering Success**

- After reading, say: “You didn’t just imagine it—you built a real version! Like Omar, your invention is now helping others speak up!”

4. **Facilitate a Reflective Discussion**

- Ask:
    - o “How do you think Omar felt pressing the button?”
    - o “What made your design work?”
    - o “Why is it important to build things that help others?”

5. **Celebrate Student Achievement**

- Applaud all students and optionally give out “Young Inventor” certificates.
- Encourage students to write thank-you notes from Omar’s perspective or draw the final scene.

Mission Innovate

Final Mission – Invent for Inclusion

Overview:

This is a culminating, open-ended project that allows students to apply all the skills they’ve gained throughout the unit. Students will design a project that helps people with hearing or speech difficulties—just like the talking glove prototype—using gestures, sounds, and smart coding. They will take full ownership of the process: from identifying a real-world challenge to building a model and presenting their solution.

Project Duration: 2 weeks

Teacher’s Role:

- **Facilitator of Inquiry:**
  Read the mission aloud and guide students in identifying places where people need help communicating. Help them focus on inclusion and empathy as they choose their challenge.
- **Encourager of Empathy and Creativity:**
  Prompt students to think about real-life difficulties—like communicating on a noisy bus or helping someone who can’t speak. Encourage thoughtful, meaningful tech-based ideas.
- **Supporter of Student-Led Work:**
  Let students take the lead in choosing a problem, planning, designing in PictoBlox, and building using craft or recycled materials. Step in only to support with reminders or scaffolded tips.

- **Guide for Reflection:**
  Remind students to revisit their earlier "Now I Can" reflections and think about what tools or strategies helped most. Use the guiding questions and checklist to support them.
- **Assessment Guide:**
  Use a rubric that covers science understanding, technology use (sensors, loops, conditionals), engineering and design quality, creativity in communication, and overall presentation.

Implementation Tips:

- Let students work in pairs or small groups to promote creativity and responsibility.
- Use a “design studio” layout—provide recycled materials, cardboard, LED/buzzer kits, and open Tinkercad or PictoBlox tabs.
- Remind them of past activities—sound training, button coding, and case design—for inspiration.
- Encourage sketching first, prototyping second.
- Celebrate all solutions that show clear thinking, even if the tech isn’t perfect.